

Predicting F1 Tyre Degradation to Optimize Pit Stop Strategy

603 Racing Group

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Problem statement

Why Tyre Degradation Prediction?

- Pit stop timing is one of the most impactful decisions in F1 racing. On average cost ~25 sec per pitstop.
- Too early: time loss from extra stops. Too late: sudden tyre performance collapse
- Teams currently rely on a mix of simulation models, gut feel, and real-time telemetry
- Goal: Build a data-driven tyre wear model to quantify degradation and recommend optimal pit stop strategies
- Success criteria: a strategist could glance at our recommendations and say “yeah, that’s a sensible call”

Dataset & data prep

Dataset

Source

FastF1 Python library — official F1 timing data

Coverage

- 2024 season and 2025 season (48 races)
- Same set of circuits across both seasons

Raw data: ~53,000 laps

Each row: lap time, compound, tyre life, stint, pit events, weather

Data cleaning

Cleaned feature	Why
In/out laps	Pit lane speed limits distort lap times, not representative of real race pace
Safety car/ Virtual safety cars/ Red flag laps	Speed is capped, not useful lap time
Wet intermediate compounds	Out of scope, model targets dry-compound degradation only
Statistical outliers	Laps > 3 std deviations from per-race mean (spins, collisions, not wear)
Starting lap	Cold-tyre chaos on first lap; excluded to keep warm-running baseline
Missing key field	Dropped rows with null: LapTime, TyreLife, Compound, or Driver

Result: 53,298 → 43,319 laps (18.7% removed)

Defining the right target

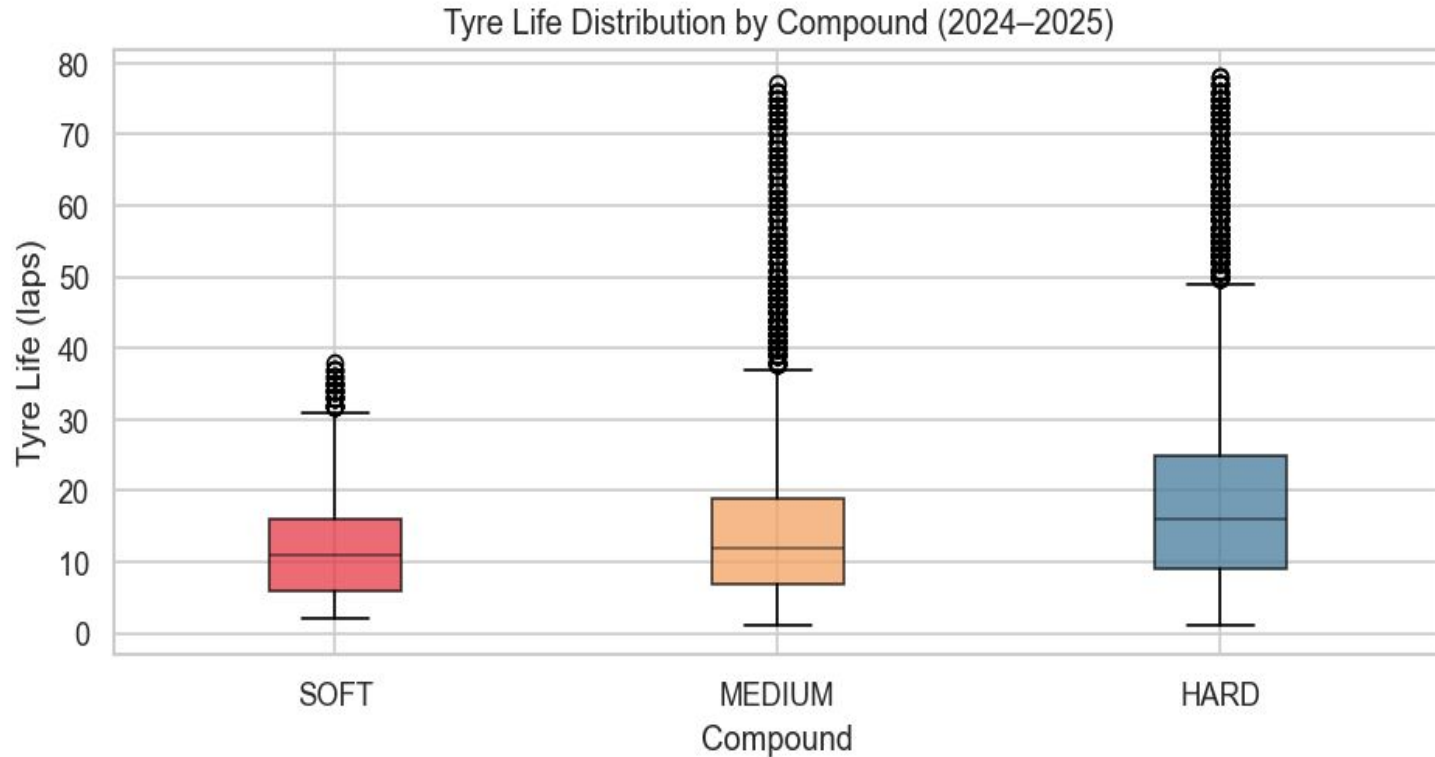
5 iterations before we landed on a usable target to predict

Iteration	Target	Result
1	Lap time minus stint median	Negative R2
2	Raw lap time	RMSE ~5s, way off
3	Lap minus first lap of stint	Negative target mean
4	Fuel-correct lap minus first lap	Cold tyre warmup biased baseline
5	Wear delta: fuel-corrected lap minus fastest fuel-corrected lap of first 5 laps of the tyre	Worked

Feature engineering

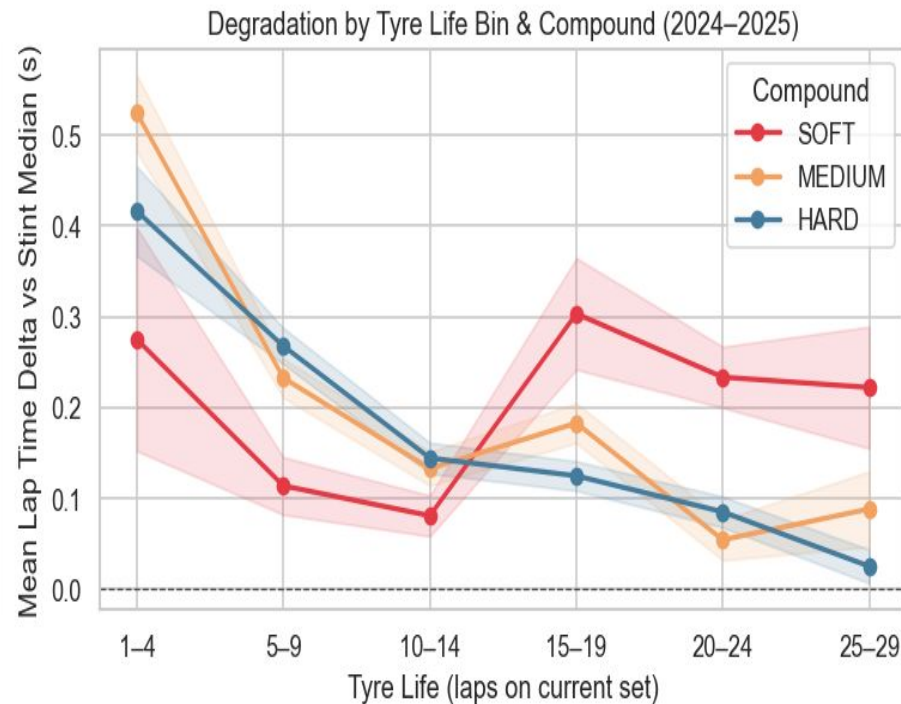
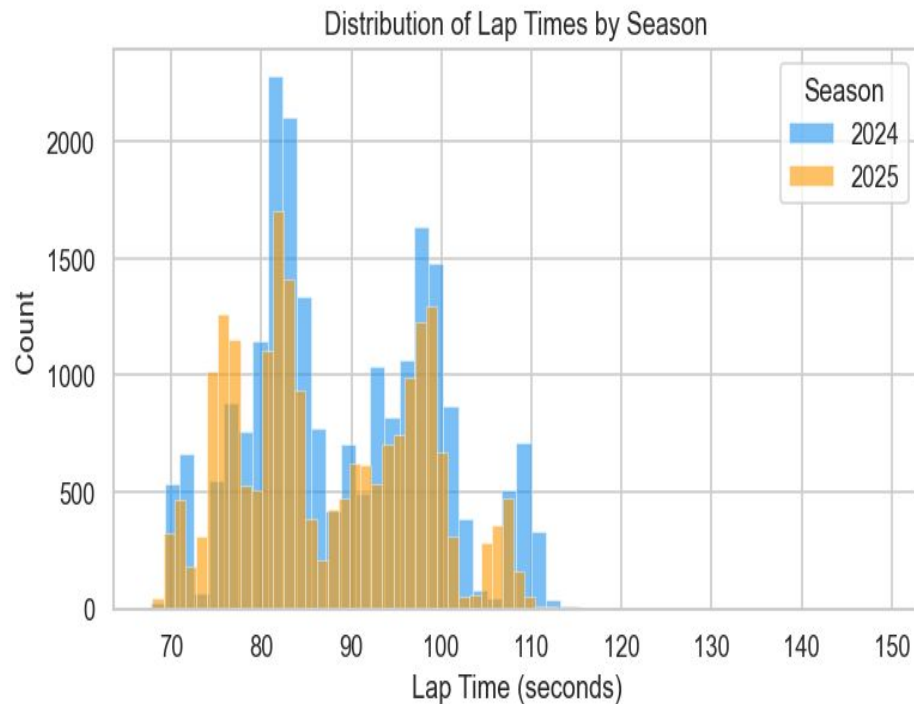
Cleaned feature	Why
WearDelta (Modeling Target)	Fuel-corrected lap time minus the fastest fuel-corrected lap in the warm up window ($\text{TyreLife} \leq 5$). Isolates pure tyre wear after stripping fuel burn-off and cold-tyre warmup.
TyreLifeRatio	TyreLife normalized 0–1 within each stint. Allows cross-stint comparison.
FuelCorrectedLap	LapTime: $0.03 \text{ s} \times \text{LapNumber}$. Removes the $\sim 0.03 \text{ s/lap}$ pace gain as fuel burns off.
CompoundEncoded	Ordinal encoding: SOFT=0, MEDIUM=1, HARD=2.
LapTimeDelta	Lap time minus per-driver per-stint median. Captures relative degradation within a stint.
StintLength	Total laps in that driver's stint. Context for strategy modeling.

EDA: Tyre Life Distribution by Compound



SOFT: short, predictable stints • MEDIUM: wide spread with long outliers • HARD: longest stints, high variability

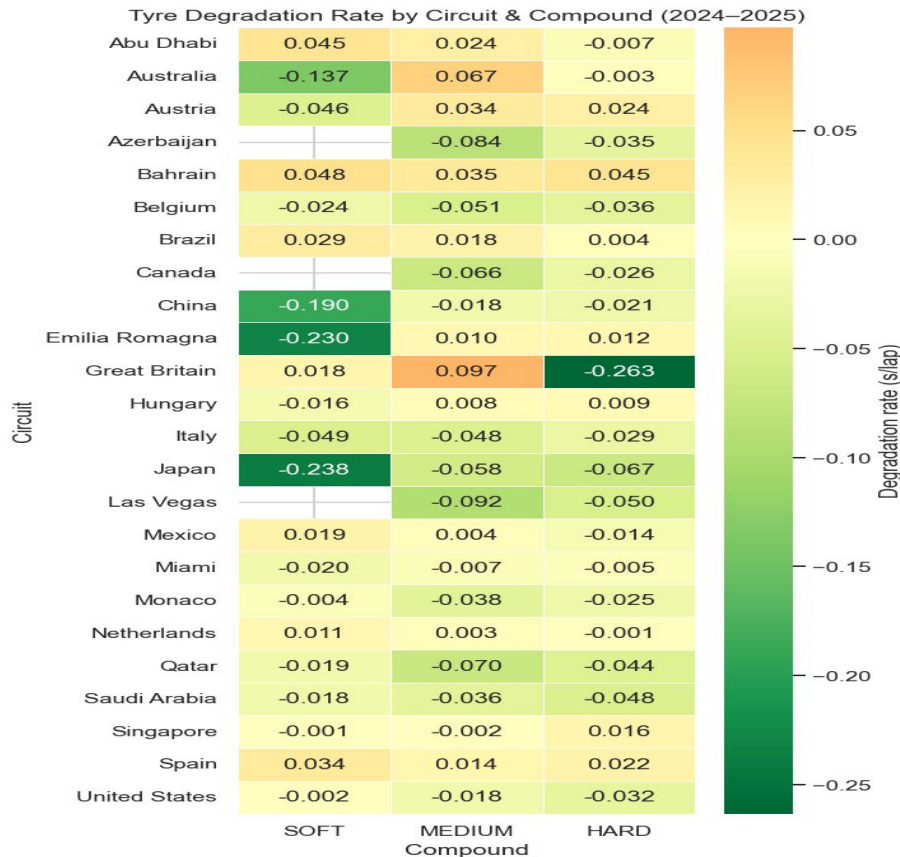
EDA: Lap Time Distributions & Degradation by Compound



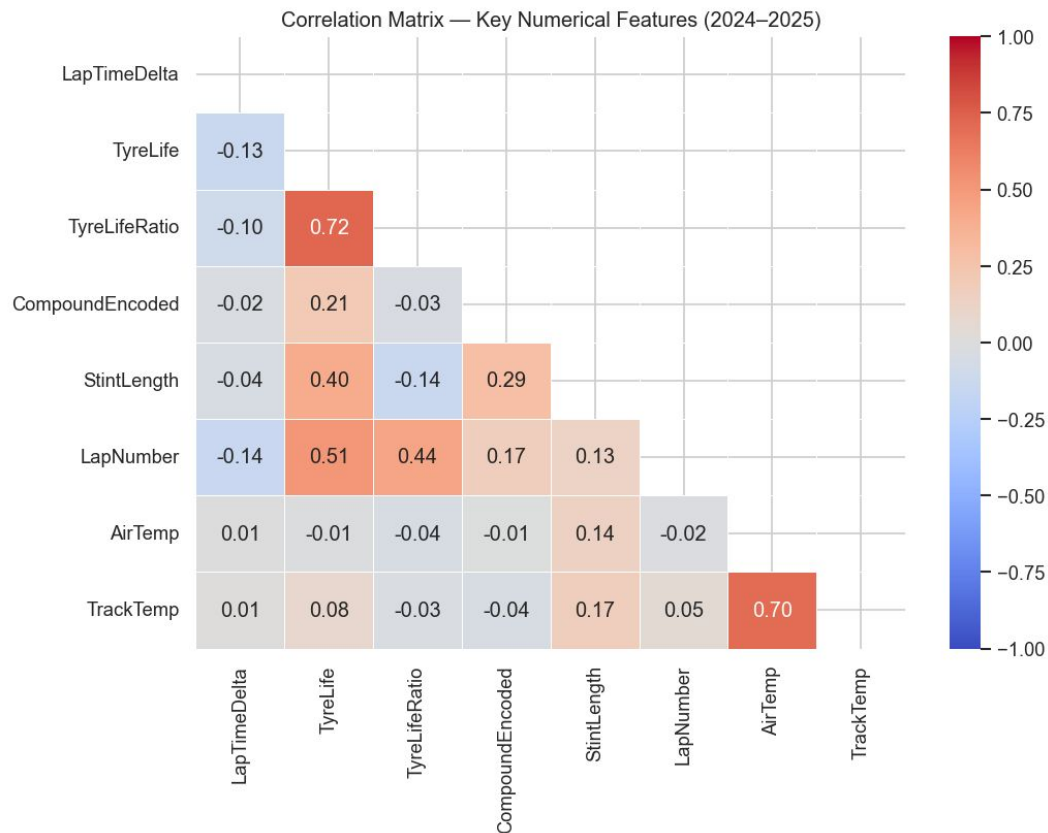
- Left: 2024 and 2025 lap time distributions nearly identical, no distribution shift in pace → safe train/test split
- Right: SOFT degrades fastest (~10 laps), MEDIUM moderate, HARD minimal over 25+ laps

EDA: Degradation Rate Heatmap — Compound × Circuit

- SOFT always degrades fastest; HARD slowest, consistent across all circuits
- High-grip circuits (Monaco, Hungary, Singapore): faster wear across all compounds
- Low-stress circuits (Monza, Spa): tyres hold pace longer
- Circuit context is essential, model needs Race as a categorical feature



EDA: Correlation Matrix — Feature Relationships



Weak linear signals

- TyreLife $\rightarrow$ LapTimeDelta: $r = 0.13$, strongest predictor, still weak
- CompoundEncoded $\rightarrow$ LapTimeDelta: $r \approx 0.03$, near-zero direct effect

Multicollinearity identified

- TyreLife $\leftrightarrow$ TyreLifeRatio: $r = 0.73 \rightarrow$ drop one
- AirTemp $\leftrightarrow$ TrackTemp: $r = 0.70 \rightarrow$ keep TrackTemp only

→ Weak linear signals point us toward non-linear tree models

Methodology

Model Setup

Target: WearDelta (seconds slower than stint's fresh-tyre pace)

Features

TyreLife, CompoundEncoded, TrackTemp, LapNumber

Features

- Linear regressions (Baseline)
- Random forest (Non-linear interactions)
- XGBoost (Winner)

Split strategy

- Initial split: Train on 2024 data, test on 2025 data
- Revised split: 80/20 at STINT level (not lap level) , prevents data leakage from partial stints

Mixed 2024 + 2025 in both train and test

32,939 laps training | 8,587 laps testing

Model performance

Model	Train RMSE	Test RMSE	Test MAE	Test R ²
Linear Regression	1.60 s	1.58 s	1.04 s	0.13
Random Forest	1.30 s	1.48 s	0.93 s	0.24
XGBoost	1.09 s	1.40 s	0.89 s	0.32

- $R^2 = 0.32$: model explains ~32% of WearDelta variance
- Tree models outperform Linear, confirming that degradation is non-linear
- Train/test RMSE gap is modest, no severe overfitting

Key results and findings

Pit-Stop optimizer strategy

For each candidate strategy (compounds + stint lengths):

1. Look up fresh-tyre pace for each compound at this circuit
2. Predict per-lap wear with XGBoost
3. Add fuel weight (cars get faster as they get lighter)
4. Add a "running on worn tyres" penalty ($0.2s \times \text{tyre age}$)
5. Add 25 seconds per pit stop
6. Sum it all up $\rightarrow$ total race time

Search every reasonable 1-stop and 2-stop strategy, pick the fastest.

Example: Bahrain

- Recommended: MEDIUM (20) $\rightarrow$ HARD (16) $\rightarrow$ MEDIUM (21)
- Predicted race time: 94.6 minutes

Pit-Stop optimizer back test (2025 season)

Vs Race winner

- Stop count match: 55% (11/20 races)
- Compound match: 55% (11/20 races [order insensitive])
- Race time within 120 sec of winner: 60% (12/20 within target range)
- Median time delta: -62 sec

Vs Top 10 finishers

- Stop count match: 80%
- Compound match: 85%

What we learned

What went right

FastF1 pipeline pulled both full seasons cleanly (~50K raw laps, ~43K after cleaning)

XGBoost captures 32% of WearDelta variance, beats Linear (13%) and Random Forest (24%) as expected.

Pit-stop optimizer turned the wear model into a concrete strategy recommender.

Optimizer recommends a strategy used by some top 10 finisher on 80% of dry races

Limitations

Per-lap noise: F1 lap times carry significant non-tyre noise: traffic, dirty air, fuel saving, driver inputs.

Pirelli compound spec shift : 2024 and 2025 used different compound specifications.

Approximate fuel correction: The 0.03 s/lap fuel correction is a rule of thumb. Actual fuel effect varies by circuit and car. Per-circuit calibration would be cleaner.

Challenges

- RMSE < 0.3 s/lap target wasn't achievable at the per-lap level: Unpredictable noise in F1 lap times (traffic, dirty air, driver inputs) makes it very difficult to achieve such accuracy.
- Took many iterations on what target to use
- Time-based train/test split didn't perform well due to compound spec changes

Future work

- **Per-circuit fuel and wear models** — calibrate fuel correction and wear curves track-by-track instead of using one global formula
- **Per-lap weather** — use lap-by-lap track temp instead of race averages

Questions?